

Hamiltonian Cycles

1 Hamiltonian Cycle Problem

Definition: Hamiltonian Cycle

Let $G = (V, E)$ be an undirected graph. A **hamiltonian cycle** is a simple cycle that contains every vertex in V exactly once and returns to the starting vertex.

A graph is called:

hamiltonian

if it contains a hamiltonian cycle. Otherwise, it is called:

nonhamiltonian

Example

In the graph below, the shaded edges form a hamiltonian cycle:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 1$$

Every vertex is visited exactly once, and the path returns to the starting vertex.

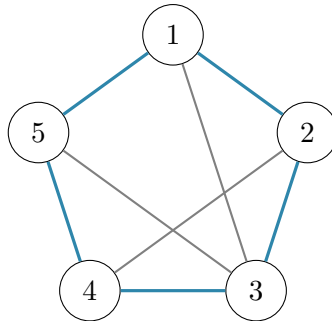


Figure: A graph with a hamiltonian cycle shown in blue.

2 Formal Language

Decision Version

The hamiltonian-cycle problem asks:

Does an undirected graph G contain a hamiltonian cycle?

As a formal language:

$$\text{HAM-CYCLE} = \{\langle G \rangle : G \text{ is a hamiltonian graph}\}$$

The input is an encoding $\langle G \rangle$ of a graph G . The output is:

Yes or **No**

3 A Nonhamiltonian Example

Bipartite Graphs with Odd Number of Vertices

Every cycle in a bipartite graph has even length because the cycle must alternate between the two vertex sets.

Therefore, if a bipartite graph has an odd number of vertices, it cannot have a hamiltonian cycle.

Reason

A hamiltonian cycle must include all vertices.

If $|V|$ is odd, then a hamiltonian cycle would have odd length. But bipartite graphs have only even-length cycles.

So the graph is nonhamiltonian.

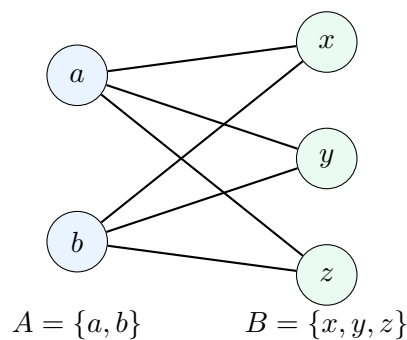


Figure: A bipartite graph with 5 vertices, so it is nonhamiltonian.

4 Naive Algorithm

Brute Force Idea

One possible algorithm lists all permutations of the vertices and checks whether any permutation forms a hamiltonian cycle.

If the graph has m vertices, then there are:

$$m!$$

possible vertex orderings.

For each ordering, the algorithm checks whether consecutive vertices are connected by edges and whether the last vertex connects back to the first.

Running Time

The brute-force algorithm is not polynomial time.

If the graph is encoded as an adjacency matrix and the encoding length is n , then:

$$m = \Omega(\sqrt{n})$$

Since there are $m!$ permutations, the running time is at least:

$$\Omega(m!) = \Omega((\sqrt{n})!)$$

This grows faster than n^k for any constant k .

5 Verification Algorithms

Definition: Verification Algorithm

A **verification algorithm** is a two-argument algorithm $A(x, y)$, where:

x = ordinary input string

y = certificate

The algorithm verifies x if there exists a certificate y such that:

$$A(x, y) = 1$$

The language verified by A is:

$$L = \{x \in \{0, 1\}^* : \text{there exists } y \in \{0, 1\}^* \text{ such that } A(x, y) = 1\}$$

6 Certificate for Hamiltonian Cycle

Certificate

For HAM-CYCLE, the certificate is a list of vertices in the order they appear in the hamiltonian cycle.

Example certificate:

$$(1, 2, 3, 4, 5)$$

The verifier checks:

1. The list is a permutation of all vertices in V .
2. Each consecutive pair is an edge in E .
3. The last vertex has an edge back to the first vertex.

Polynomial Verification

Given graph G and certificate y , the verifier can check the proposed cycle in polynomial time. Using a simple implementation, the verification can run in:

$$\mathcal{O}(n^2)$$

where $n = |\langle G \rangle|$ is the length of the graph encoding.

7 Why Ham-Cycle is in NP

Membership in NP

A problem is in **NP** if every **Yes** instance has a certificate that can be verified in polynomial time.

For HAM-CYCLE:

- If G is hamiltonian, a hamiltonian cycle gives a valid certificate.
- If G is nonhamiltonian, no certificate can make the verifier accept.
- The verifier checks the certificate in polynomial time.

Conclusion

$$\text{HAM-CYCLE} \in \mathbf{NP}$$

In fact, the hamiltonian-cycle problem is **NP-complete**, but proving NP-completeness requires a polynomial-time reduction.

8 Key Ideas

Term	Meaning
Hamiltonian cycle	A simple cycle that visits every vertex exactly once and returns to the start.
Hamiltonian graph	A graph that contains at least one hamiltonian cycle.
Nonhamiltonian graph	A graph with no hamiltonian cycle.
Certificate	A proposed ordering of all vertices in the cycle.
Verifier	Checks whether the certificate is a valid hamiltonian cycle.
Ham-Cycle	The language of all encoded hamiltonian graphs.